

Dual-Map Spatial Augmentation for Collaborative VR Crisis Management

Hugo Lino*

Ana Rita Rebelo†

Rui Nóbrega‡

Fernando Birra§

Nuno Correia¶

NOVA-LINCS

NOVA School of Science and Technology, Lisbon, Portugal

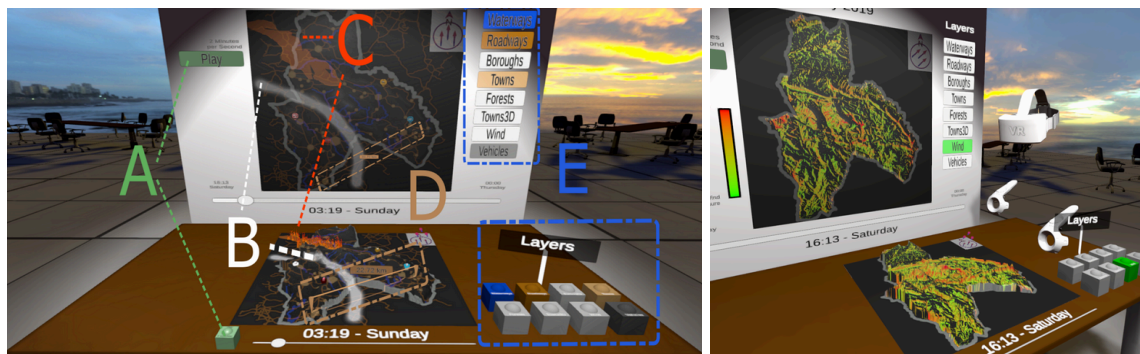


Figure 1: View of the dual map interface in virtual reality (VR). The left side shows an overview of system features including a play button for wildfire playback (A), brush strokes from the brush tool (B), wildfire visualization (C), and a measurement made with the ruler tool (D). Geographic information layers (E) are activated for waterways, roadways, towns, and vehicles. The right side illustrates a second user and the wind visualization layer with wind blowing from the southwest to the northeast.

ABSTRACT

Geospatial data visualization in crisis management is traditionally confined to 2D displays, which limits users' ability to intuitively grasp complex topographical data related to disaster progression. This research explores the concept of spatial augmentation by introducing a dual-map interface within a collaborative virtual reality (VR) environment designed to support joint analysis, communication, and skill development. The system synchronizes a vertical 2D wall map for high-level coordination with a horizontal table map for tactical 3D terrain analysis, allowing multiple users to share spatial understanding and coordinate decisions in real time. We conducted a user study ($N = 18$) using data from a real-world wildfire scenario to evaluate how spatial orientation influences user preference and accuracy across the two mapping visualizations and interaction modalities. Our analysis reveals a statistically significant performance advantage for the table interface in spatial measurement accuracy, despite a steeper initial learning curve. Additionally, we identify a correlation between participant height and interface preference, with shorter users showing some reluctance toward the fixed-height table interface. These findings suggest complementary benefits for both interfaces and indicate that collaborative VR-based spatial augmentation tools can enhance communication and teamwork in augmented environments, but must account for user ergonomics to be effective.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

*e-mail:h.lino@alumni.fct.unl.pt

†e-mail:ar.rebelo@campus.fct.unl.pt

‡e-mail:rui.nobrega@fct.unl.pt

§e-mail:fpb@fct.unl.pt

¶e-mail:nmc@fct.unl.pt

1 INTRODUCTION

Collaborative crisis management is a cognitively demanding domain that requires the rapid synthesis of multi-layered geospatial information, including terrain elevation, weather patterns, and shifting resource locations. In high-pressure environments such as wildfire suppression, the ability to maintain a Common Operational Picture (COP) is essential for survival and strategic success. Traditional situation rooms use the spatial organization of the physical environment, employing vertical whiteboards for high-level overviews and horizontal tables for collaborative path planning and troop movement coordination. Virtual reality (VR) enables these physical metaphors to be augmented through real-time data synchronization and immersive 3D topography, which would be impractical or impossible to achieve with physical hardware alone.

However, it remains unclear how the spatial orientation of these interfaces affects human performance. Most existing VR mapping research focuses on either a tabletop view [12] or a vertical display [4] in isolation. We propose a dual-map spatial augmentation approach in which both interfaces coexist within a shared virtual space and are synchronized in real time. This configuration mirrors the collaborative structure of a physical command center while offering the benefits of a 3D terrain visualization on the table and a 2D situational overview on the wall. Our study investigates whether these interfaces are redundant or complementary, and examines how human physical characteristics, such as height, influence interaction and performance within augmented spatial environments.

2 RELATED WORK

The availability of geospatial information is a fundamental requirement for effective crisis management and response [7,9]. Prior research has explored the use of digital tabletops to support shared situational awareness among response teams [2]. This concept was later adapted to VR, where interfaces are positioned on horizontal surfaces to support immersive decision making [12]. These collaborative environments aim to establish a common operational picture by enabling multiple users to analyze geospatial data together within a shared virtual space.

VR has also become an established tool for training in complex emergency scenarios. Applications have been developed for a wide range of tasks, including firefighter training [14] and mass-casualty incident management [1]. Other systems focus on close-protection serious games [8] or strategic training for operational managers [6]. These applications emphasize the importance of realistic simulations for enhancing human performance and skill acquisition. By immersing users in high-stakes environments, such systems provide a safe space for practitioners to develop the cognitive and sensory capabilities required for real-world crisis response.

Despite these advances, the specific spatial arrangement of user interfaces in VR requires further investigation especially in collaboration settings [5, 10, 13]. Designing augmented mapping interfaces for VR involves a trade-off between egocentric and exocentric viewpoints. While some studies have compared VR-based mapping with traditional desktop interfaces [3] or investigated different projections and perspectives for global maps [4, 11, 15], there is limited research on how users transition between interfaces with different spatial orientations. In particular, few studies directly compare vertical and horizontal configurations within a single virtual environment. This work addresses this gap by examining how a dual-map configuration can extend users' spatial interaction capabilities and support a more flexible collaborative environment.

3 DUAL-MAP INTERFACE FOR CRISIS MANAGEMENT

The primary objective of the system is to provide an augmented situation room in which multiple users can share geospatial information and coordinate responses to a crisis. The environment is built around the concept of spatial augmentation, employing a dual-map interface to provide complementary perspectives on a wildfire scenario (Figure 1). By combining a vertical wall map with a horizontal table map, the system augments the physical layout of a situation room with interactive, synchronized data layers.

The core of the system is the dual-map configuration. The wall map (Figure 2a) is a 2D vertical interface that provides a high-level overview of the crisis area, making it well suited for identifying landmarks and monitoring large-scale events. Its vertical orientation and elevated placement allow the information to remain easily visible to multiple users at a distance, supporting shared awareness across the room. In contrast, the table map (Figure 2) is a 3D horizontal interface that uses heightmap data to visualize terrain elevation and supports hands-on interaction with objects. The horizontal layout enables intuitive spatial manipulation and provides haptic feedback during direct interaction, facilitating precise tasks such as measurement and annotation. Together, these interfaces support the simultaneous exploration of the same geospatial data from complementary perspectives within a shared virtual environment.

All interactions within the system are synchronized in real time. When a participant toggles a data layer or performs a measurement on one interface, the change is immediately reflected on the other map. This synchronization ensures that all users maintain a common

operational picture, regardless of their physical position in the room or their preferred interaction surface.

3.1 Interaction Features

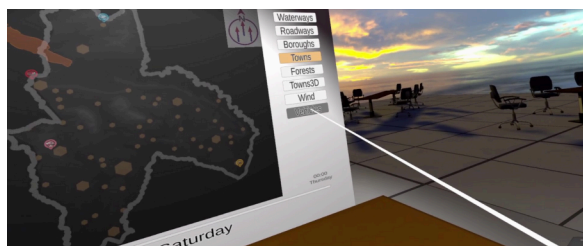
The system provides a set of synchronized interaction tools designed to support seamless use across both map orientations. Users can toggle eight different data *layers*, highlighted in blue in Figure 1, including waterways, road networks, boroughs, towns, forests, wind data, and vehicle locations, using diegetic buttons positioned adjacent to each map. Interaction techniques are adapted to the spatial context of each interface: the wall map employs a ray-casting pointing metaphor, while the table map supports proximal interaction, providing haptic and visual feedback when users reach out to touch the virtual surface.

The dynamic data layers support risk assessment and decision making. A wildfire playback visualization (Figure 1-(A)) enables users to observe the temporal progression of the fire, while a wind visualization tool illustrates how wind direction interacts with terrain slope and aspect. To enhance situational awareness, the terrain is augmented with color-coded risk indicators. Red regions represent high-risk zones where wind and terrain conditions may accelerate fire spread, while green areas indicate more sheltered locations. By projecting these risk factors directly onto the three-dimensional terrain of the table map, the system transforms abstract geospatial variables into immediate visual cues, supporting more informed predictions of fire behavior and resource deployment strategies.

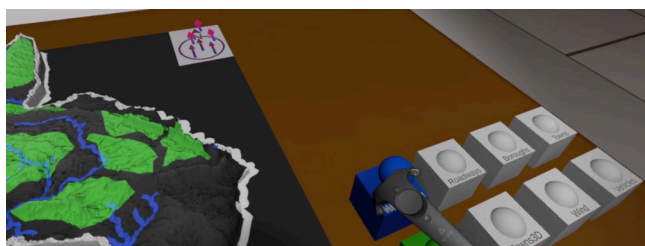
To support strategic analysis, the system includes a *ruler tool* (Figure 3) for distance measurement and a pressure-sensitive *brush*



Figure 3: Distance measurements performed on the table map using the ruler tool, synchronized and reflected on the wall map interface.



(a) User toggling the Vehicle layer, using the wall interface.



(b) User toggling the Vehicle layer, using the table interface.

Figure 2: User toggling the Waterways layer, using the two interfaces.

tool (Figure 1-(B)) for freehand annotation. These tools enable participants to delineate exclusion zones, annotate areas of interest, and measure distances between critical elements, such as the distance between a fire front and nearby settlements.

To facilitate team coordination, we implemented an *action proposal mechanism*. When a user proposes a resource movement (e.g., vehicle movement), all other participants receive a haptic notification accompanied by a visual indicator. This allows the team to review the proposal on either map interface and collaboratively approve or reject the action. By anchoring these interactions to the dual-map configuration, the system supports shared decision making while keeping relevant geospatial information continuously visible.

3.2 Technical Setup

The system was developed using the Unity Game Engine and is designed for room-scale VR using the HTC Vive Pro headset. To ensure spatial consistency across both map interfaces, we employed the EPSG:3857 coordinate reference system. Geospatial data are sourced from OpenStreetMap¹ for infrastructure, along with high-resolution heightmap data for terrain rendering.

Collaborative functionality is implemented using the Mirror² networking framework, which synchronizes user avatars, interface states, and interactions across the network. For tactical path planning, the system integrates the Open Source Routing Machine to compute realistic vehicle movements based on the underlying road network.

4 USER STUDY

To evaluate how spatial orientation influences interaction behaviour, performance, and user preference in the proposed dual-map environment, we conducted a user study using data from a real-world wildfire scenario given by Mação Municipality, Portugal. The study was designed to assess both objective performance measures and subjective user responses when interacting with vertical and horizontal map interfaces. In particular, we examined interface tendency, preference, task completion time, and spatial measurement accuracy, while also investigating the influence of users' physical characteristics on interaction outcomes.

4.1 Demographics

The study included a total of $N = 18$ participants, with ages ranging from 12 to 32 years (Mdn = 24). Regarding prior experience with VR, the participant pool was evenly split, with 9 participants reporting previous VR experience and 9 reporting no prior exposure. Almost all participants ($n = 17$) identified as regular video game players, and 2 participants reported prior experience with professional emergency management systems.

The mean participant height was 1.70 ± 0.09 meters, with 11 participants taller than the mean and 7 shorter. This physical characteristic was recorded to examine how user height interacted with the fixed dimensions of the virtual furniture. The table map was positioned at a height of 0.79 m and spanned a 1.0 m square surface, while the wall map was placed 2.84 m from the user's initial position and occupied a 2.0 m square area starting at 0.71 m above the floor.

4.2 Task Design

Four primary metrics were used to evaluate interaction within the dual-map environment: tendency, preference, completion time, and absolute error. *Tendency* was defined as the user's natural inclination to select one interface over the other when no explicit instruction was provided. This metric was measured in Task Groups A and D by recording which map participants interacted with first. *Preference* captured users' self-reported interface choice and was collected

¹OpenStreetMap: <https://www.openstreetmap.org/> (09/01/26)

²Mirror Networking: <https://mirror-networking.com/> (09/01/26)

Table 1: Summary of the task groups.

Group	Task Objectives
A: Exploration	A1 - Activate the Waterways and Roadways layers. A2 - Activate the Wind layer and interact with it. A3 - Find Amêndoa and Cardigos towns on the upper half of the map
B: Layer Interaction	B1 - Activate the Vehicle layer (using wall interface). B2 - Activate the Forests layer (using wall interface). B3 - Deactivate the Vehicle layer (using table interface). B4 - Deactivate the Forest layer (using table interface).
C: Measurement	C1/C3 - Distance between Cardigos and the red truck. (Ruler Tool) C2/C4 - Distance between Amêndoa and the blue truck
D: Annotation (Brush Tool)	D1 - Highlight a specific town (Interface of user's choice)

through a post-task questionnaire after participants completed identical tasks using both map orientations (Task Groups B and C). *Completion time* corresponded to the total duration, in seconds, required to successfully complete each task and was automatically logged by the system. *Absolute error* was evaluated exclusively for ruler tool tasks and quantified spatial accuracy as the difference, measured in virtual kilometers, between the distance estimated by the participant and the ground-truth distance obtained from the Geographic Information System (GIS) dataset.

The study followed a structured workflow composed of four task groups, summarized in Table 1. Task Group A (Exploration) required participants to identify landmarks and activate weather-related data layers in order to assess initial orientation behavior. Task Group B (Layer Interaction) involved toggling vehicle and forest datasets first on the wall interface and subsequently on the table interface, enabling a comparison of interaction efficiency across orientations. Task Group C (Measurement) required participants to compute distances between wildfire fronts and nearby towns using the ruler tool. Finally, Task Group D (Annotation) allowed participants to highlight a region of interest using the interface of their choice.

5 RESULTS

The collected data provide insight into how spatial orientation influences user behavior, performance, and preference within the proposed dual-map environment.

Tendency and Learning Effects: During the initial acclimation phase (Task Group A), we observed a clear familiarity bias. Approximately 78% of participants initially chose to interact with the vertical wall map (Figure 4a). This tendency suggests that users initially project prior experience with 2D computer screens and physical whiteboards onto the virtual environment.

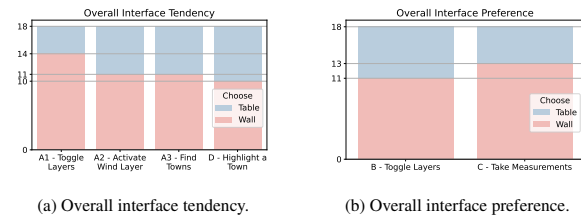


Figure 4: Participants' overall interface tendency and preference.

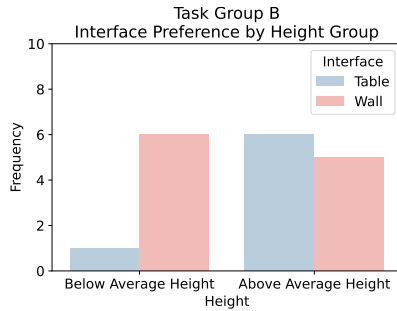


Figure 5: Participants' interface preference for task B, grouped by height.

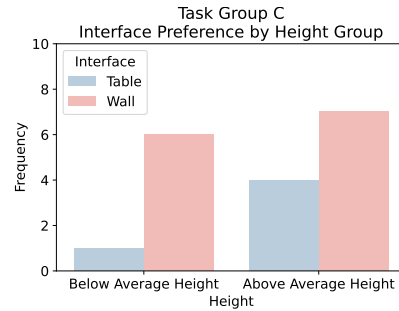


Figure 6: Participants' interface preference for task C, grouped by height.

This effect was also reflected in the completion time data for Task Group B. For first-time interactions, the wall interface was significantly faster ($M = 2.90$ s, $SD = 0.90$ s) than the table interface ($M = 6.18$ s, $SD = 5.39$ s). A paired-sample t-test confirmed this difference to be statistically significant ($t(17) = -2.57, p = 0.02$).

However, this performance advantage was not observed in the second interaction. By the second iteration of the same task, performance on the table interface improved substantially ($M = 3.16$ s), reaching statistical parity with the wall interface. This result indicates that while vertical interfaces may better support initial orientation, horizontal spatial interfaces exhibit a rapid learning curve, with users quickly adapting to direct 3D interaction.

Measurement Accuracy: In Task Group C (Measurement), task completion times were comparable across interfaces. However, map orientation had a significant impact on measurement accuracy. When using the table interface, participants achieved a significantly lower absolute error ($M = 0.49$ km) than when using the wall interface ($M = 0.58$ km). A Welch two-sample t-test showed a statistically significant difference ($t(17) = 2.35, p = 0.03$).

The horizontal orientation of the table interface allowed participants to lean forward, positioning their eyes and hands within close proximity to the data points being measured. In contrast, interaction with the wall interface required ray-casting from a distance of approximately 3 meters to maintain full visibility of the map. At this distance, small involuntary hand movements are amplified, resulting in increased angular error. The table interface therefore enhances fine motor precision by enabling grounded, near-field interaction with spatial data rather than relying on distant pointing.

User Preference and Ergonomic Factors: Subjective interface preference was divided among participants (Figure 4b). Further analysis revealed that this split was influenced by participant height. As shown in Figures 5 and 6, participants below the mean height of 1.70 m demonstrated a statistically significant preference for the wall interface ($p < 0.05$ across Task Groups B and C). Qualitative feedback indicated that the fixed table height of 0.79 m was less ergonomically suitable for shorter users, who reported needing to reach uncomfortably across the virtual surface to access the center of the map. In contrast, taller participants described the table interface as more "tactile" and "natural" for tactical tasks. These findings highlight that the effectiveness of spatial augmentation is strongly influenced by anthropometric alignment between users and virtual interface elements.

6 DISCUSSION

The results of this study highlight a trade-off in spatial augmentation: the interface that provides the highest objective accuracy is not necessarily the one users prefer. Although the tabletop interface revealed significantly more accurate measurements, the majority

of users (72%) still preferred the wall interface. We attribute this preference to perspective and familiarity. The 2D wall provides a consistent and orthographic view that users perceive as stable regardless of their height, whereas the 3D table requires users to adjust their physical posture continuously to maintain a clear line of sight.

The observed effects of user height and the initial learning curve ($p = 0.02$) point toward important design considerations for future VR augmentation tools. Shorter users' aversion to the table interface suggests that fixed spatial layouts can limit accessibility. For a system to effectively augment human capabilities, virtual interfaces should allow adjustments in height and tilt based on users' anthropometric data. If the table height had been adjustable, it is likely that the performance gains in accuracy would have been accompanied by a corresponding increase in user preference.

Moreover, the data indicate that the two interfaces are complementary rather than redundant. The wall interface serves as a gateway interface, supporting rapid orientation and initial task acquisition, while the table interface functions as a specialist interface, essential for tasks requiring high spatial precision. In collaborative crisis scenarios, this duality enables a division of labor: one user can manage high-level coordination on the wall map while another performs precise terrain analysis on the table, thereby enhancing the collective efficiency of the team.

7 CONCLUSION AND FUTURE WORK

This research demonstrates that a dual-map interface provides a robust foundation for VR-based crisis management by leveraging the complementary strengths of vertical and horizontal spatial orientations within a collaborative immersive environment. Our study shows that while the horizontal tabletop interface significantly improves spatial measurement accuracy, users often find vertical interfaces more immediately approachable, particularly for shared orientation and communication. Importantly, we identify human physical height as a significant predictor of interface preference, underscoring the need for ergonomically adaptive VR systems.

Future work will focus on implementing adaptive spatial layouts that dynamically scale and reposition virtual furniture according to users' physical dimensions and interaction context. We also plan to evaluate the system in high-stress, multi-user scenarios to better understand how spatial redundancy and role-based interaction support communication, teamwork, and skill development over time. By tailoring collaborative augmented environments to both physical and cognitive user needs, we aim to contribute to the design of more effective, inclusive, and scalable VR tools for crisis management training and research.

ACKNOWLEDGMENTS

This research was supported by NOVA LINCS (UID/04516/2025) and grant FCT/IP (2022.12085.BD).

REFERENCES

- [1] H. Berndt, D. Wessel, L. Willer, M. Herczeg, and T. Mentler. Immersion and presence in virtual reality training for mass casualty incidents. 05 2018.
- [2] S. Doeweling, T. Tahiri, P. Sowinski, B. Schmidt, and M. Khalilbeigi. Support for collaborative situation analysis and planning in crisis management teams using interactive tabletops. In *Proceedings of the 2013 ACM International Conference on Interactive Tabletops and Surfaces, ITS '13*, p. 273–282. Association for Computing Machinery, New York, NY, USA, 2013. doi: 10.1145/2512349.2512823
- [3] W. Dong, T. Yang, H. Liao, and L. Meng. How does map use differ in virtual reality and desktop-based environments? *International Journal of Digital Earth*, 13(12):1484–1503, 2020. doi: 10.1080/17538947.2020.1731617
- [4] I. Giannopoulos, A. Komninos, and J. Garofalakis. Natural interaction with large map interfaces in vr. In *Proceedings of the 21st Pan-Hellenic Conference on Informatics, PCI 2017*. Association for Computing Machinery, New York, NY, USA, 2017. doi: 10.1145/3139367.3139424
- [5] A. Giovannelli, L. Pavanatto, S. Davari, H. Miao, V. Chheang, B. Giera, T. Bremer, and D. A. Bowman. Investigating the influence of playback interactivity during guided tours for asynchronous collaboration in virtual reality. In *2025 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*, pp. 23–33, 2025. doi: 10.1109/VR59515.2025.00027
- [6] J. Himmelstein, A. Ahmad, O. Balet, J.-B. de la Rivière, M. Schaap, W. Overdijk, E. Gobetti, G. Pintore, F. Ganovelli, and P. Brivio. Interactive simulation technology for crisis management and training: The indigo project. p. 144, 04 2012.
- [7] H. Lino, R. Nóbrega, F. Birra, and N. Correia. Managing a crisis in virtual reality - tackling a wildfire. In *Proceedings of the 27th ACM Symposium on Virtual Reality Software and Technology, VRST '21*. Association for Computing Machinery, New York, NY, USA, 2021. doi: 10.1145/3489849.3489957
- [8] H. Lukosch, T. Van Ruijven, and A. Verbraeck. The other city - designing a serious game for crisis training in close protection. In *ISCRAM 2012 Conference Proceedings - 9th International Conference on Information Systems for Crisis Response and Management, ISCRAM 2012 Conference Proceedings - 9th International Conference on Information Systems for Crisis Response and Management*. Simon Fraser University, Jan. 2012. 9th International Conference on Information Systems for Crisis Response and Management, ISCRAM 2012 ; Conference date: 22-04-2012 Through 25-04-2012.
- [9] R. Nóbrega, A. Sabino, A. Rodrigues, and N. Correia. Flood emergency interaction and visualization system. In M. Sebillio, G. Vitiello, and G. Schaefer, eds., *Visual Information Systems. Web-Based Visual Information Search and Management*, pp. 68–79. Springer Berlin Heidelberg, Berlin, Heidelberg, 2008.
- [10] A. R. Rebelo, P. A. Ferreira, and R. Nóbrega. Techniques for multiple room connection in virtual reality: Walking within small physical spaces. *IEEE Transactions on Visualization and Computer Graphics*, 31(5):2310–2319, 2025. doi: 10.1109/TVCG.2025.3549895
- [11] A. Santos-Torres, T. Zarraonandia, P. Díaz, and I. Aedo. Comparing visual representations of collaborative map interfaces for immersive virtual environments. *IEEE Access*, 10:55136–55150, 2022. doi: 10.1109/ACCESS.2022.3176949
- [12] Y. Sermet and I. Demir. An immersive decision support system for disaster response. In *26th ACM Symposium on Virtual Reality Software and Technology, VRST '20*. Association for Computing Machinery, New York, NY, USA, 2020. doi: 10.1145/3385956.3422087
- [13] X. Wang, L. Shen, L. Chen, M. Fan, and L. Lee. Teamportal: Exploring virtual reality collaboration through shared and manipulating parallel views. *IEEE Transactions on Visualization and Computer Graphics*, 31(5):3314–3324, 2025. doi: 10.1109/TVCG.2025.3549569
- [14] C. H. Wijkmark, I. Heldal, and M.-M. Metallinou. Experiencing immersive vr simulation for firefighter skills training. In A. Adrot, R. Grace, K. Moore, and C. W. Zobel, eds., *ISCRAM 2021 Conference Proceedings – 18th International Conference on Information Systems for Crisis Response and Management*, pp. 913–921. Virginia Tech, Blacksburg, VA (USA), 2021.
- [15] Y. Yang, B. Jenny, T. Dwyer, K. Marriott, H. Chen, and M. Cordeil.

Maps and globes in virtual reality. *Computer Graphics Forum*, 37(3):427–438, 2018. doi: 10.1111/cgf.13431